

POLITECNICO
MILANO 1863

Vibration control with a piezoelectric bender with an active electric impedance

Group 9

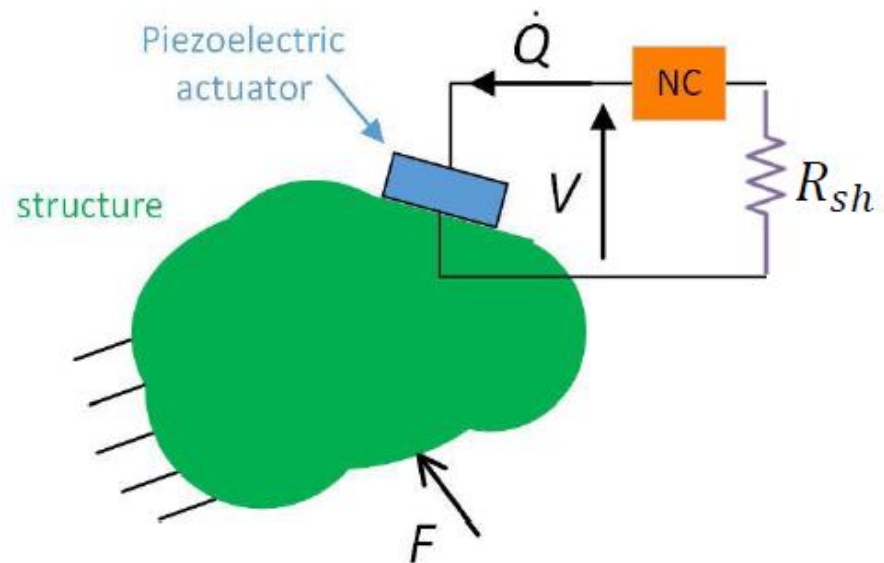
Riccardo Frasson Christian Galli Edoardo Giacon
Giorgio Morra Sara Orazzo

Project Introduction

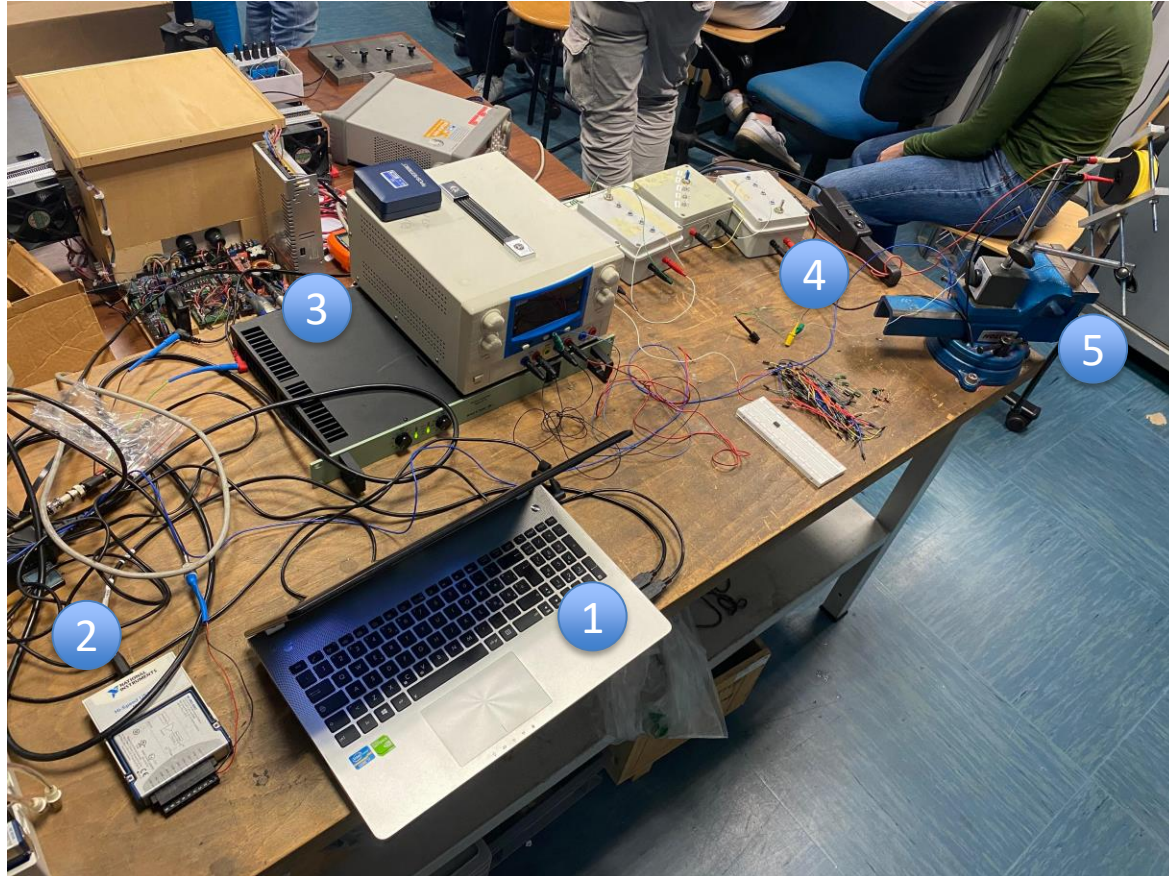
Project's Aim: Perform vibration control on a simple structure by shunting a piezoelectric bender with an active electric impedance

Procedure:

- Derive optimal layout for the shunt impedance
- Make tests without the shunt impedance
- Build the shunt impedance and make tests with the shunt impedance
- Modal extraction
- Compare experimental and theoretical results



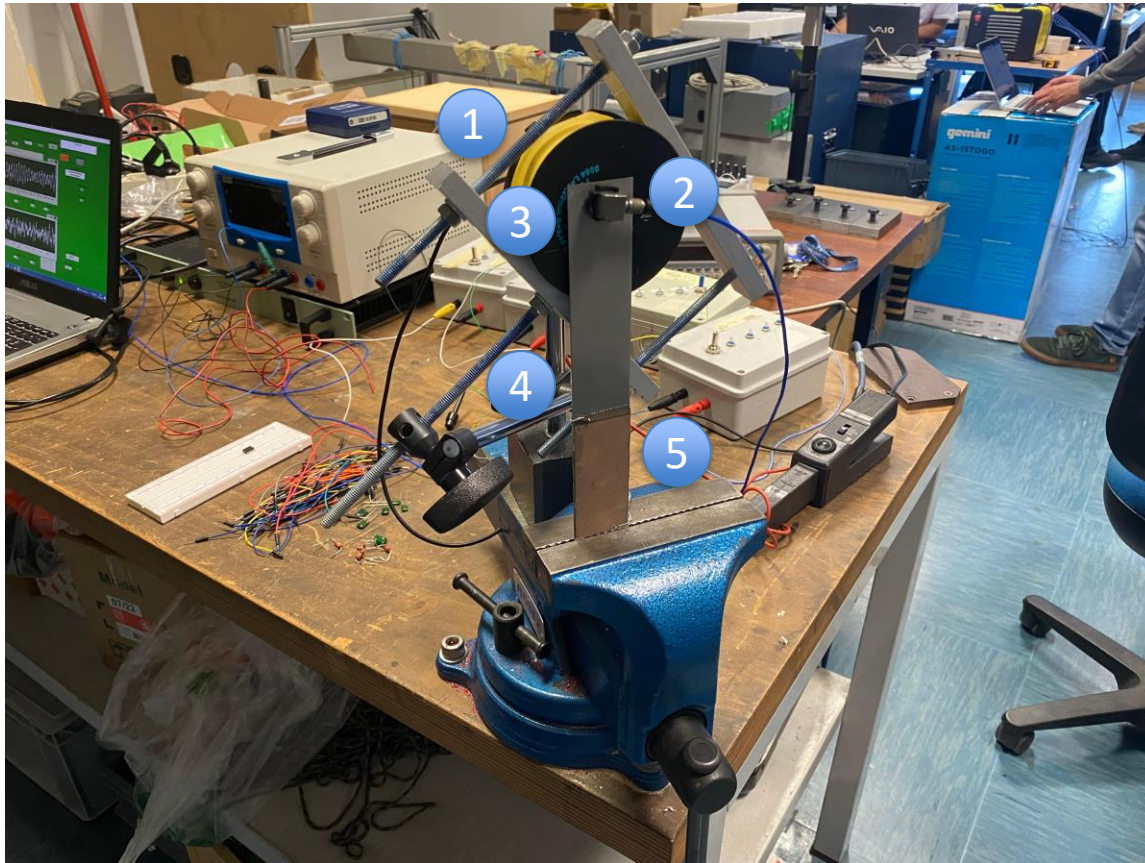
Measurement chain set up



Components:

1. Signal Generator (Pc)
2. DAC
3. Amplifier
4. Current Sensor
5. Mechanical system

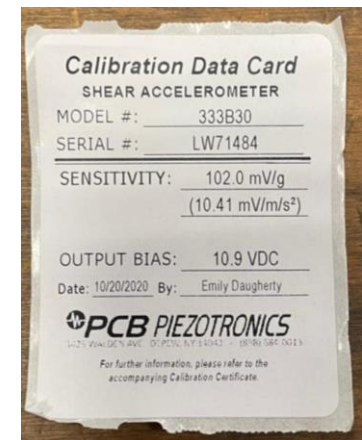
Measurement chain set up – cont'd



Mechanical System

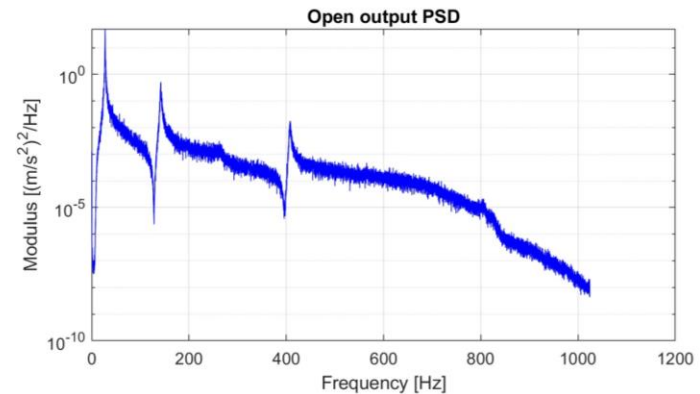
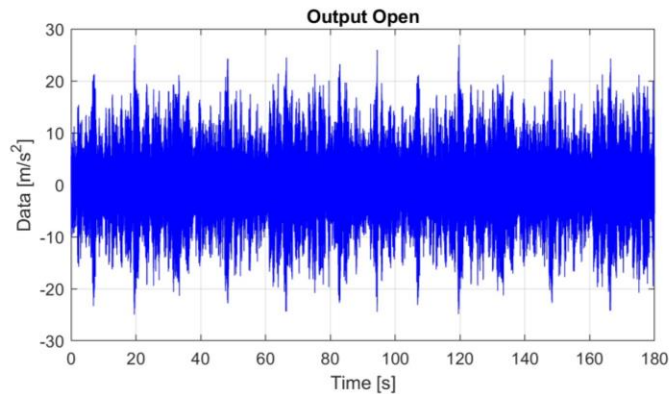
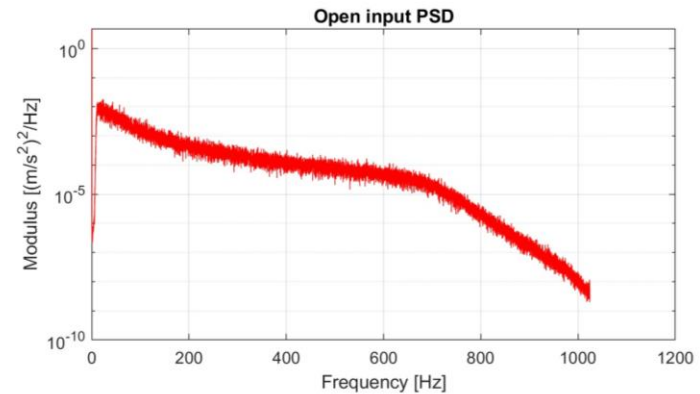
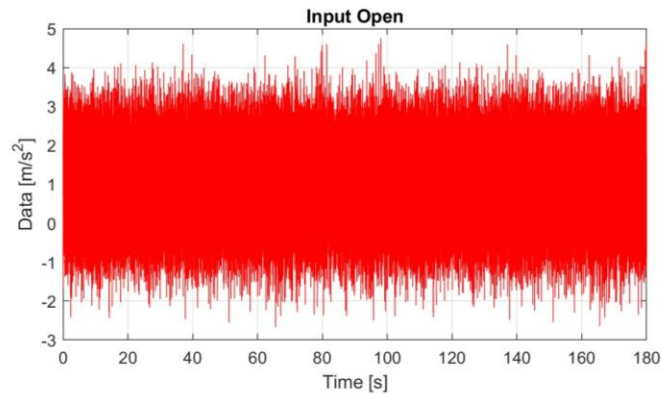
Components:

1. Magnetic Actuator
2. Accelerometer
3. Permanent magnet
4. Cantilever Beam
5. Piezoelectric Patch



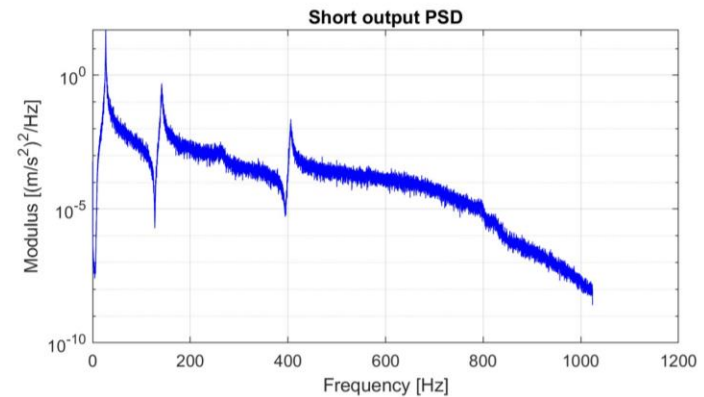
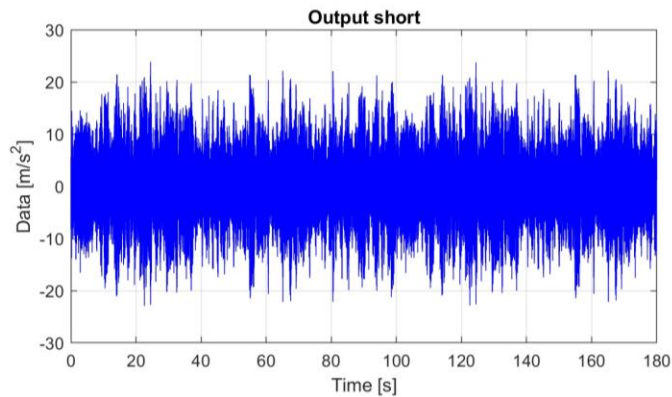
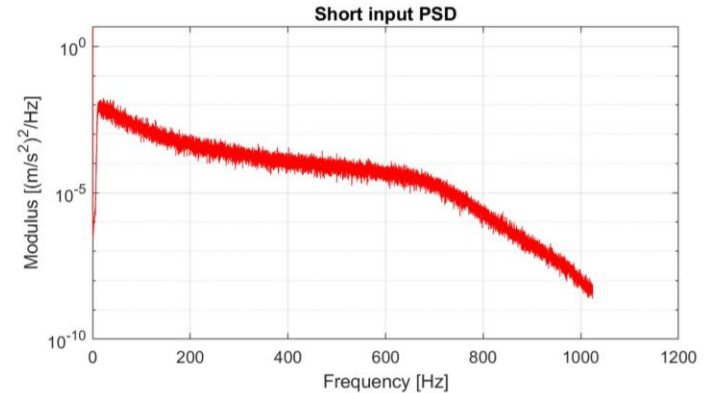
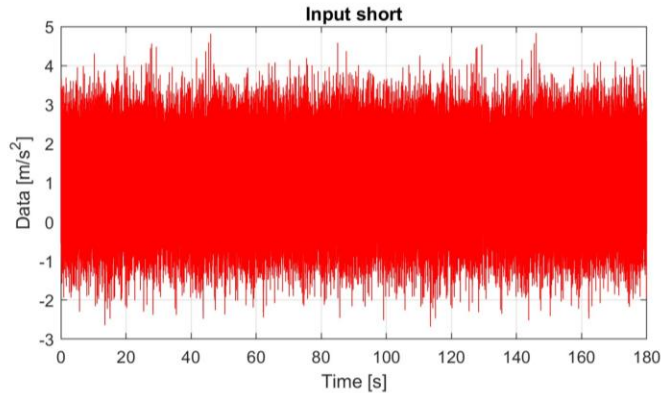
Acquired Data – System's Response (Open circuit configuration)

Acquired data

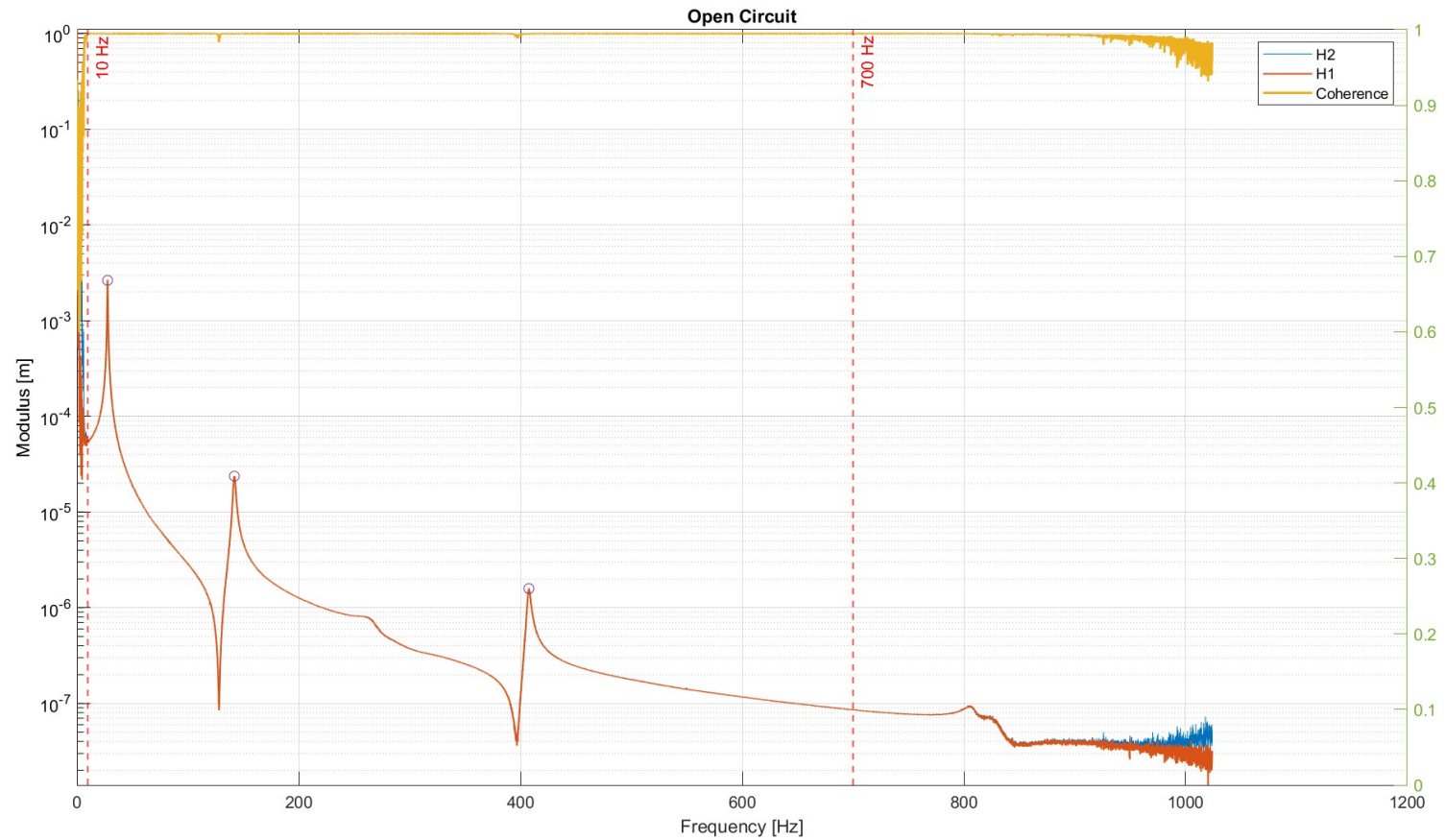


Acquired Data – System's Response (Short circuit configuration)

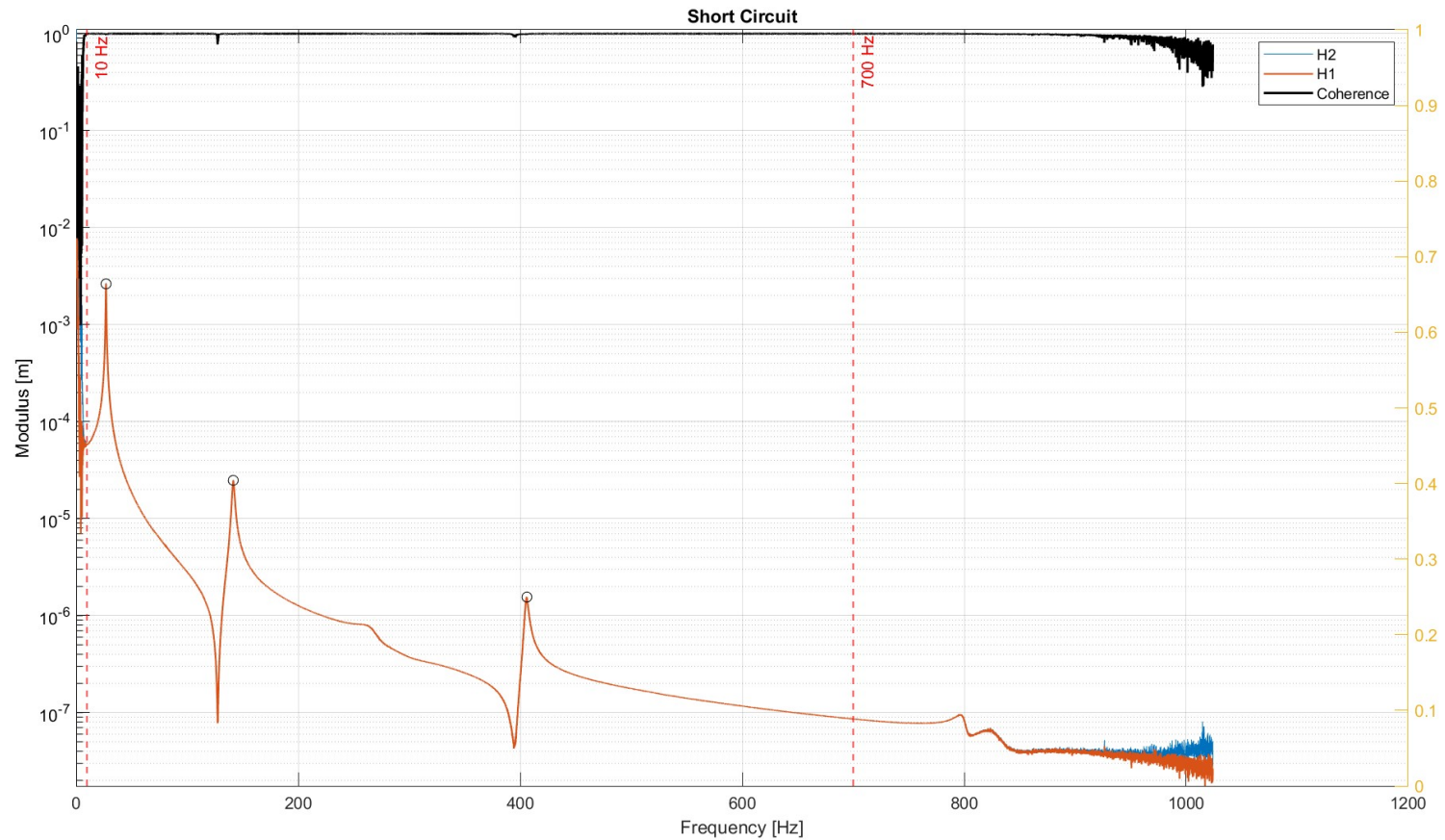
Acquired data



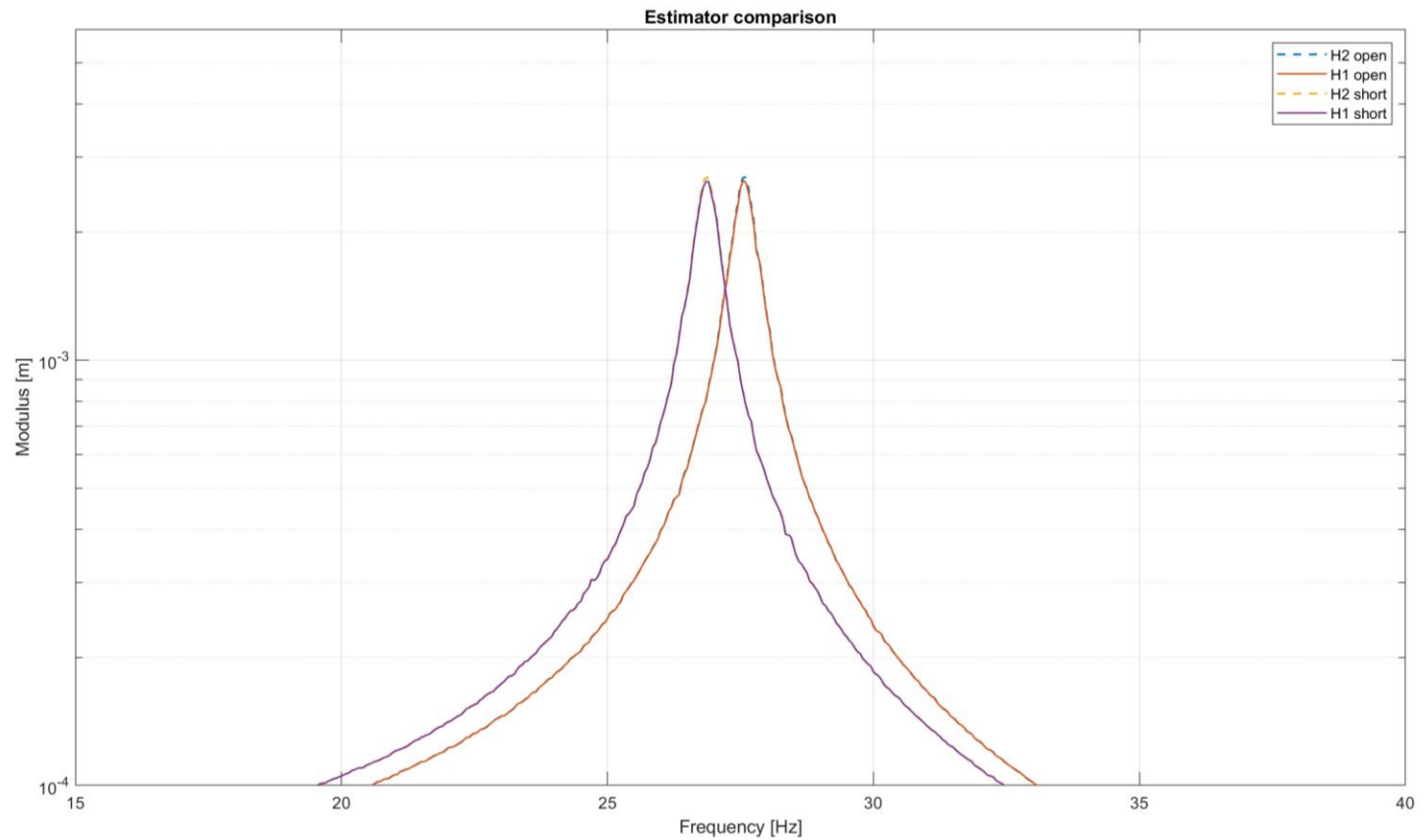
FRF Estimation – Open Circuit



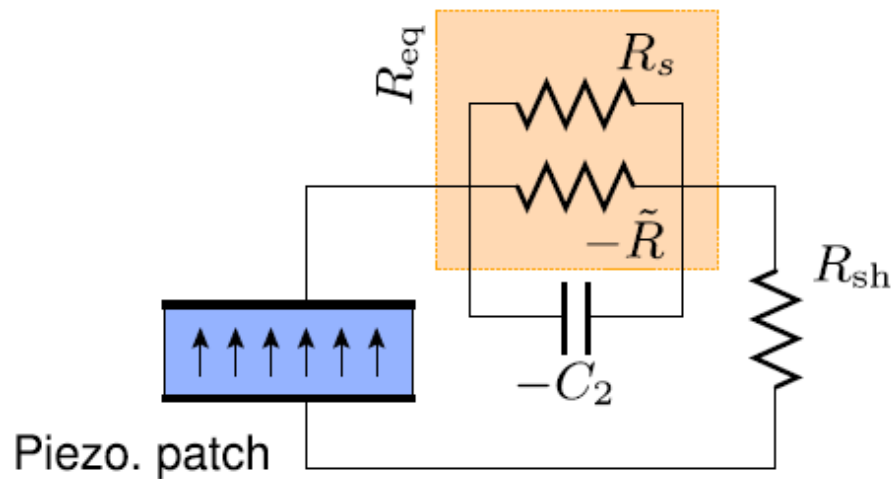
FRF Estimation – Short Circuit



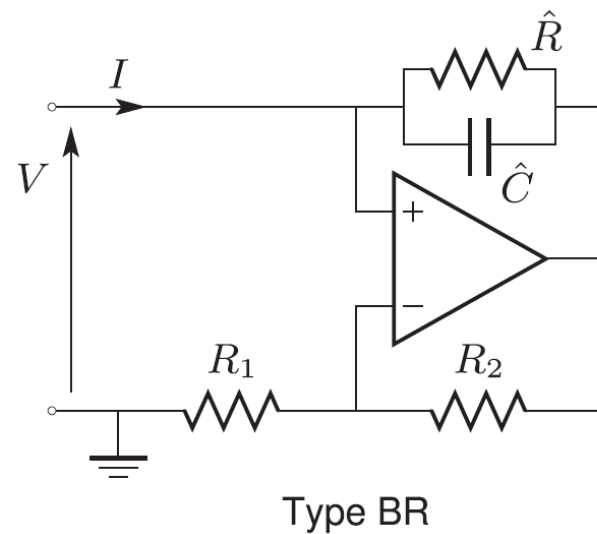
FRF Estimation – Overall Comparison



Circuit layout

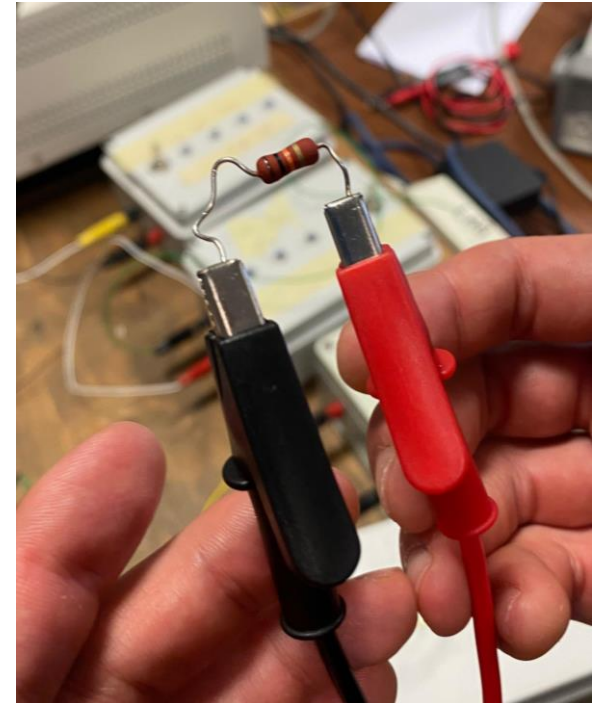
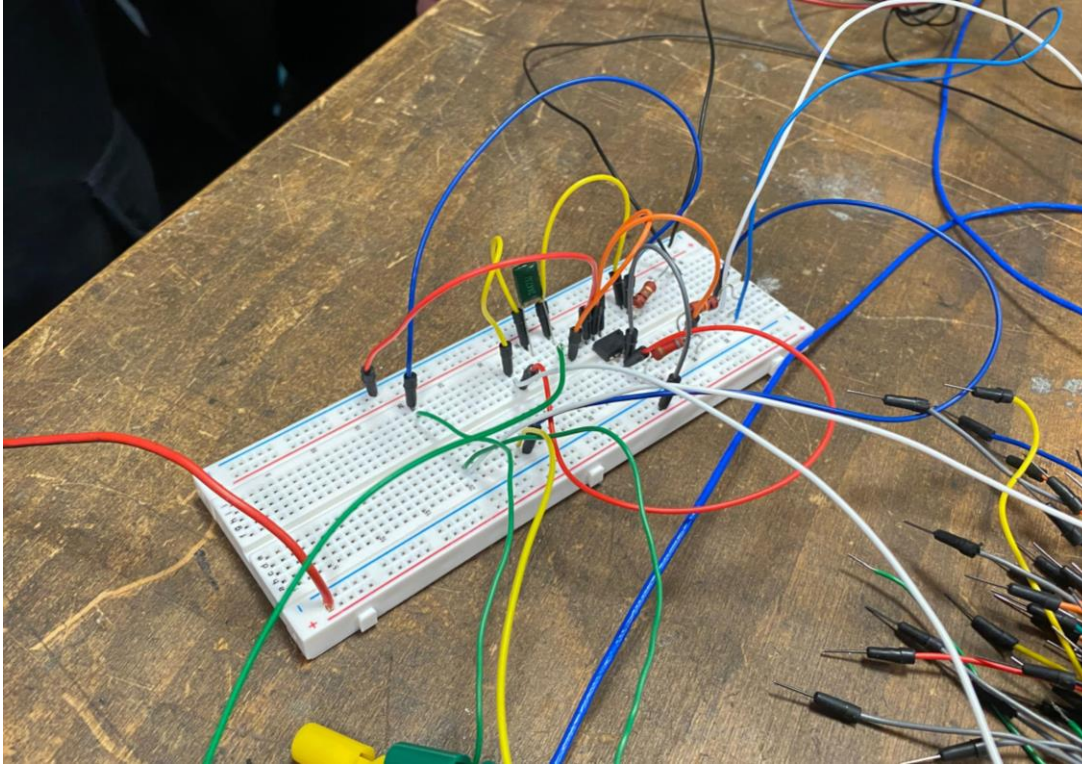


Complete circuit



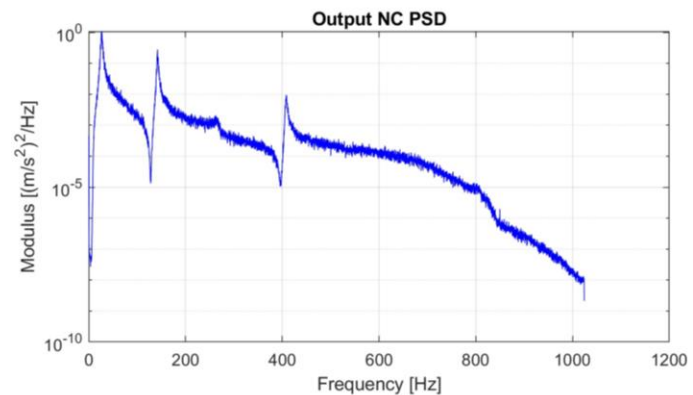
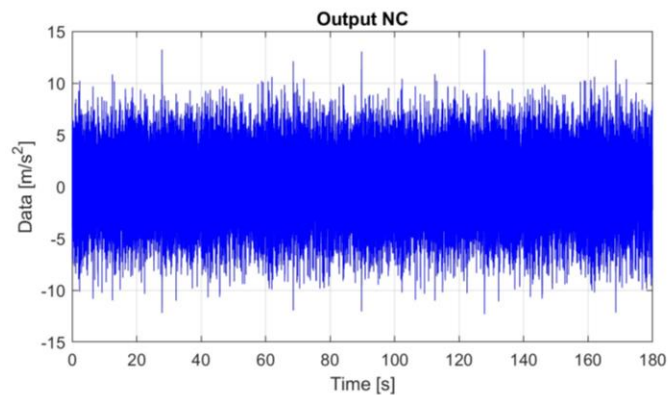
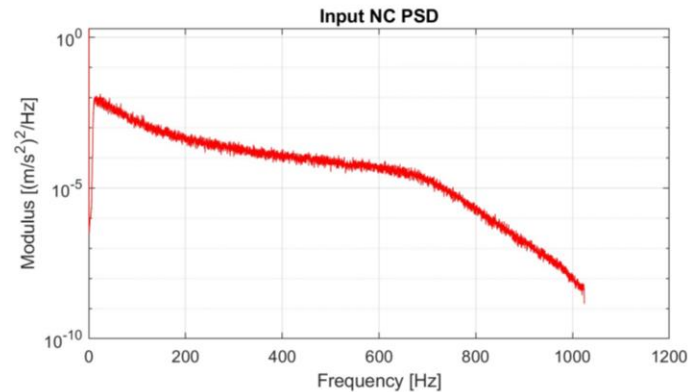
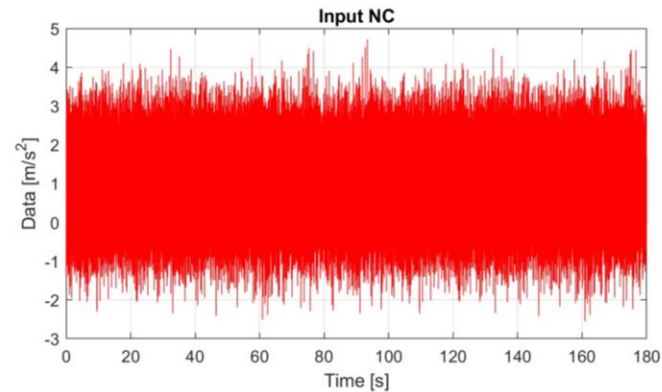
Negative Capacitance

Circuit layout – cont'd

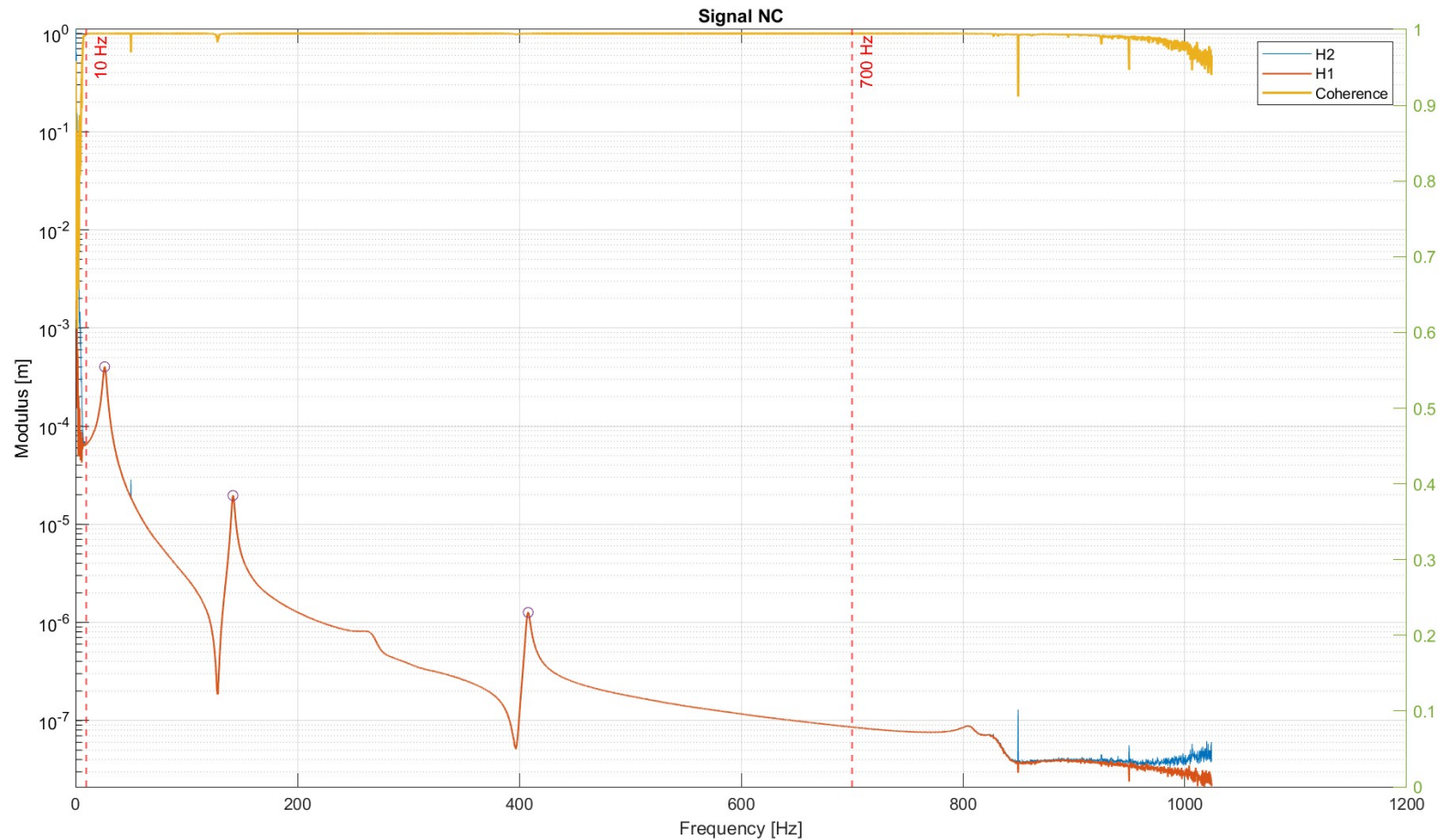


Vibration Control – NC in series configuration (Acquired Data)

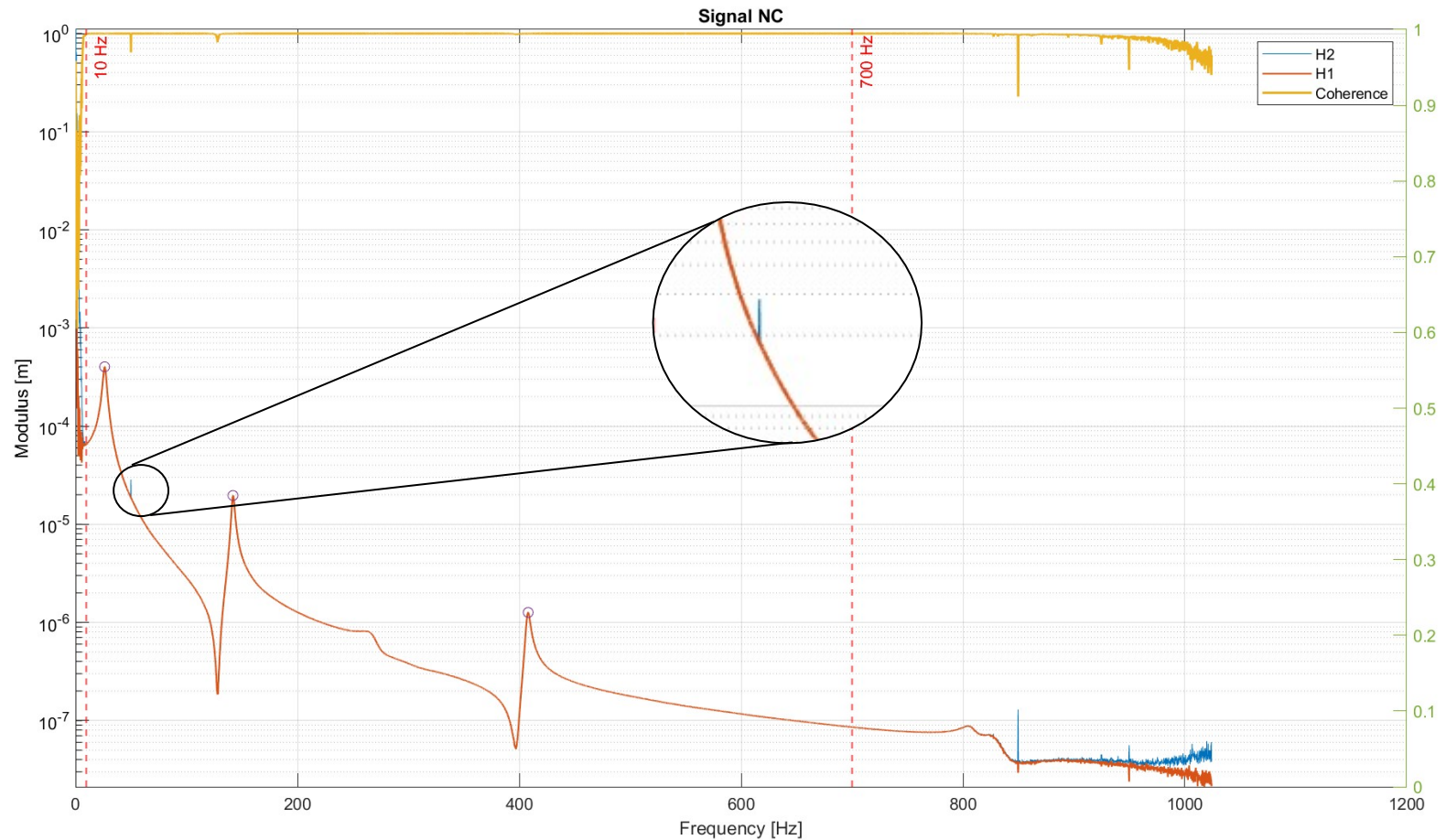
Acquired data



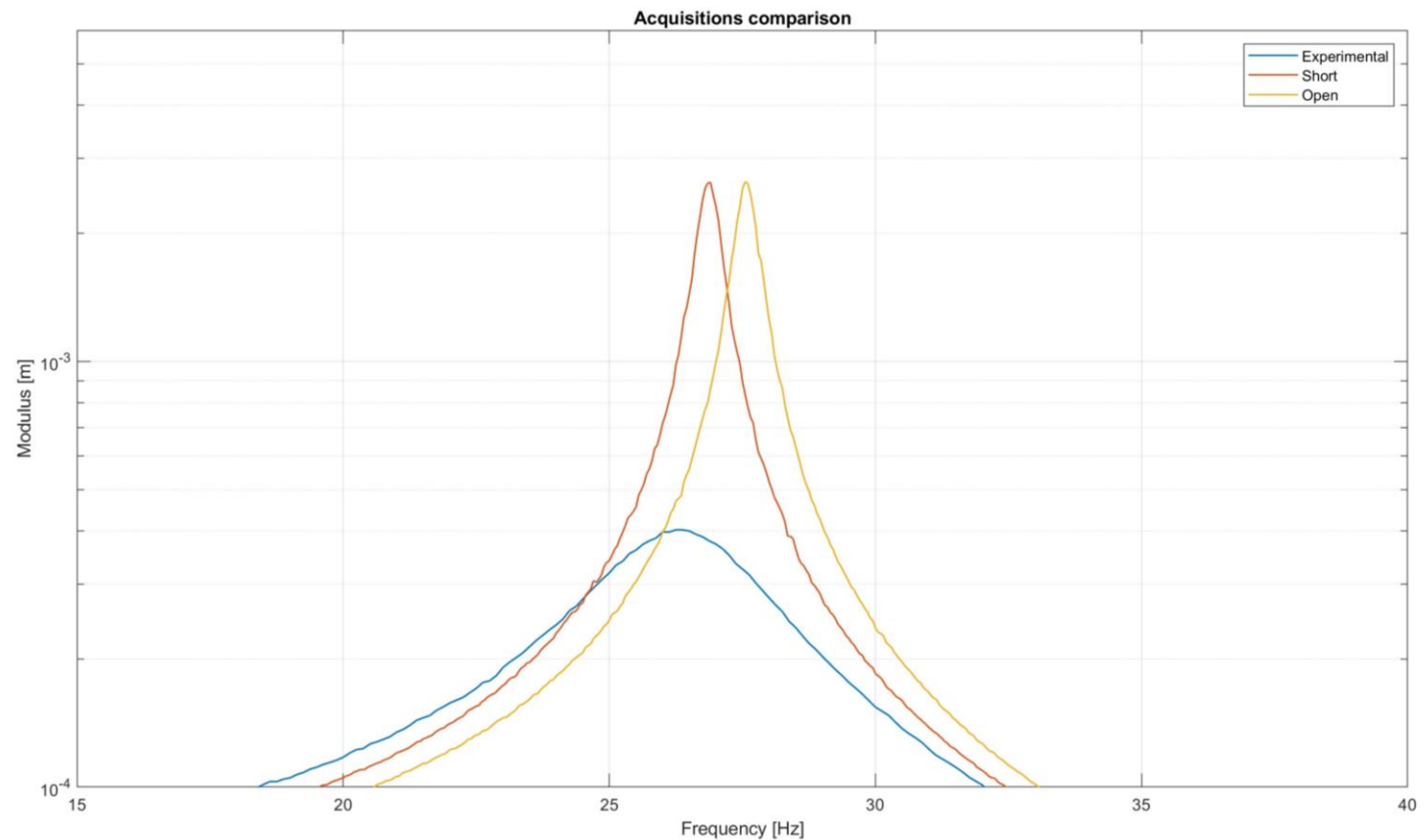
Vibration Control – NC in series configuration (FRF Estimation)



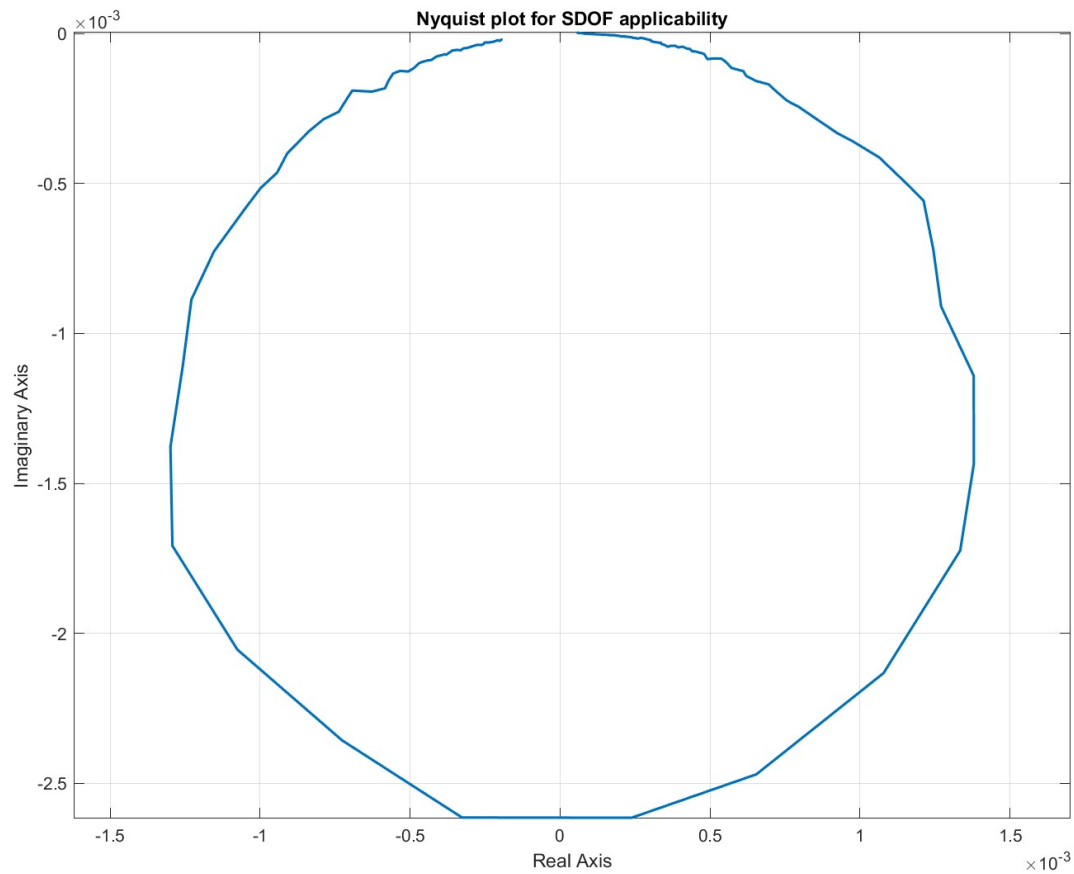
Vibration Control – NC in series configuration (Non-linearity)



Vibration Control – Overall Comparison



Optimal Values Estimation – SDOF Validation



Optimal Values Estimation – Modal Extraction (Short circuit)

Analytical model:

$$\frac{A_{jk}^{(r)}}{\omega_r^2 - \omega^2 + 2i\xi_r\omega\omega_r} + \frac{B_{jk}^{(r)}}{\omega^2} + C_{jk}^{(r)}$$

Error Function: $\min \sum_{k=m}^n [(Re_{exp}(\omega_k) - Re_{th}(\omega_k))^2 + (Im_{exp}(\omega_k) - Im_{th}(\omega_k))^2]$

First guesses:

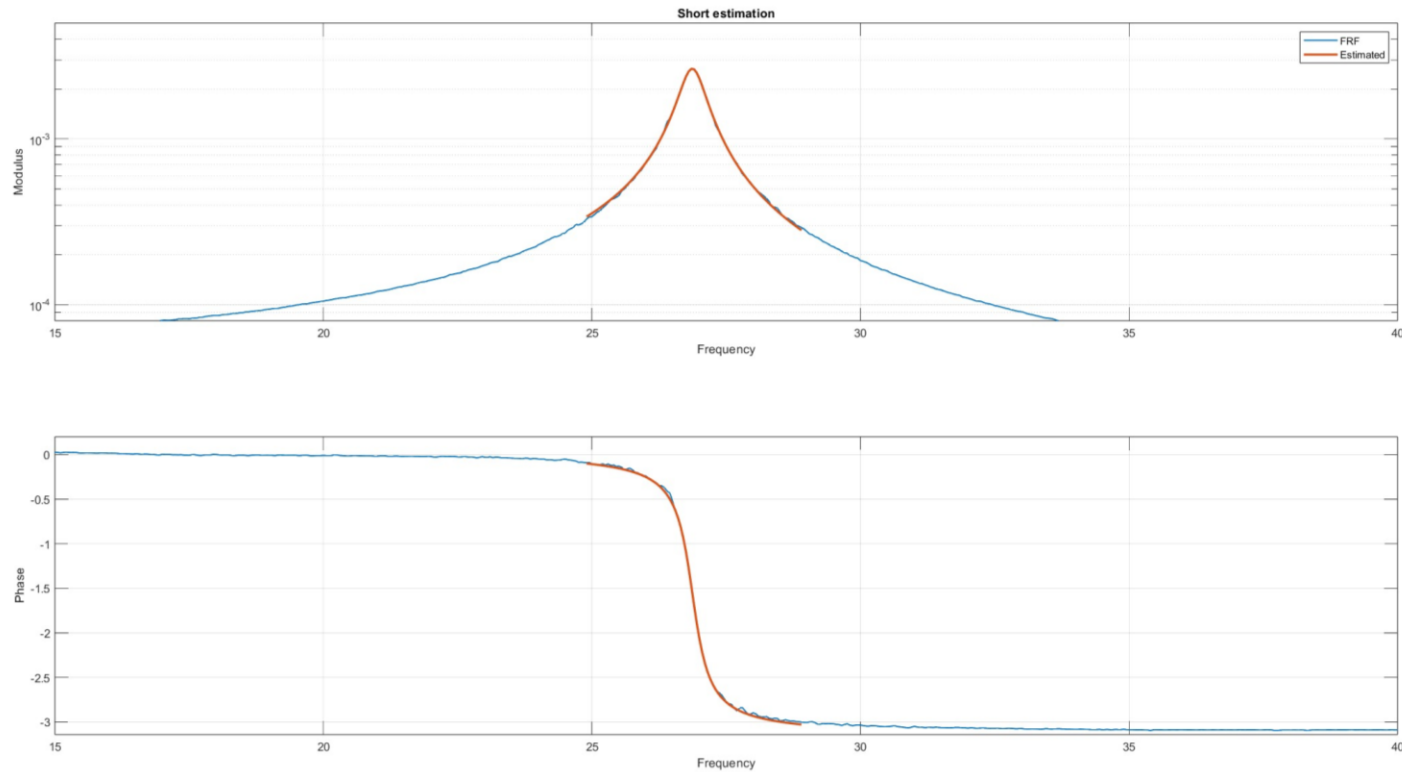
ω_0 : Findpeaks on matlab

$A_{jk}^{(0)}$: Experimental FRF Magnitude

ζ_0 : Phase derivative method

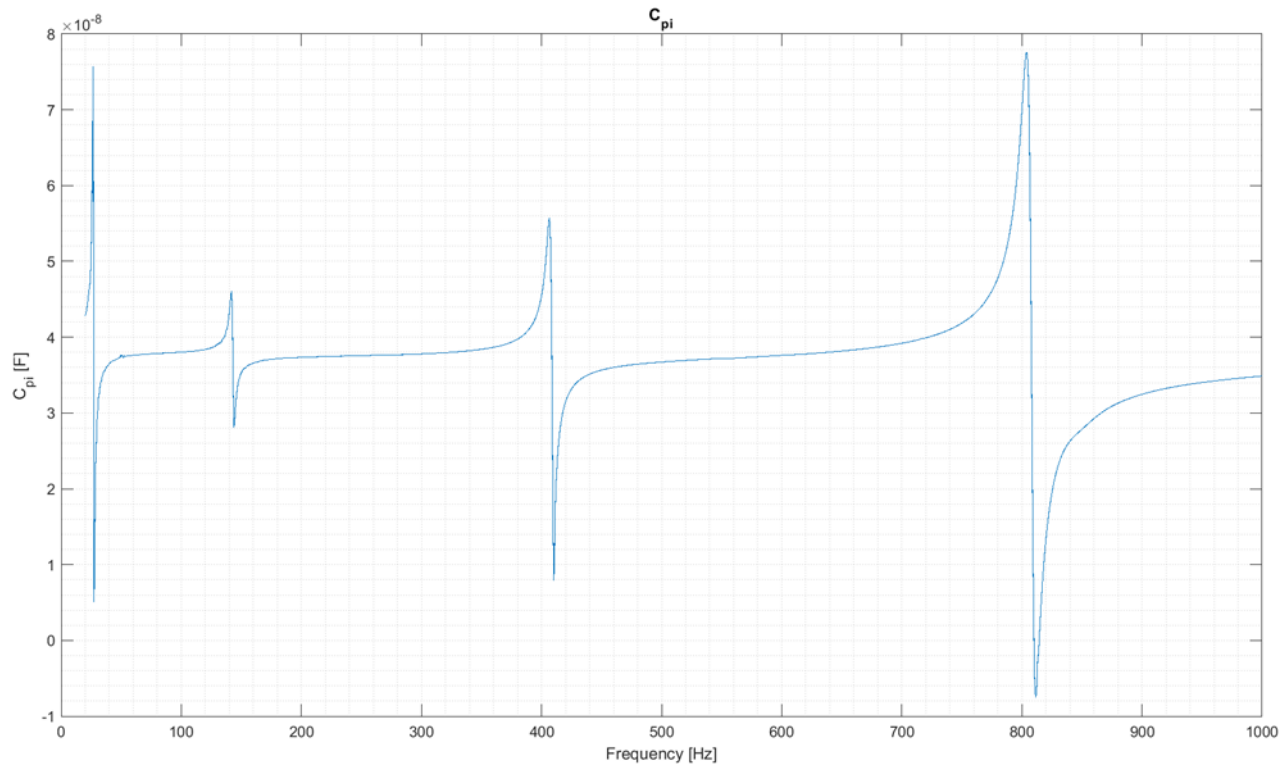
$B_{jk}^{(0)}, C_{jk}^{(0)}$: Null Contribution

Optimal Values Estimation – Modal Extraction (Short circuit)



$$\xi = 0.88 \%$$

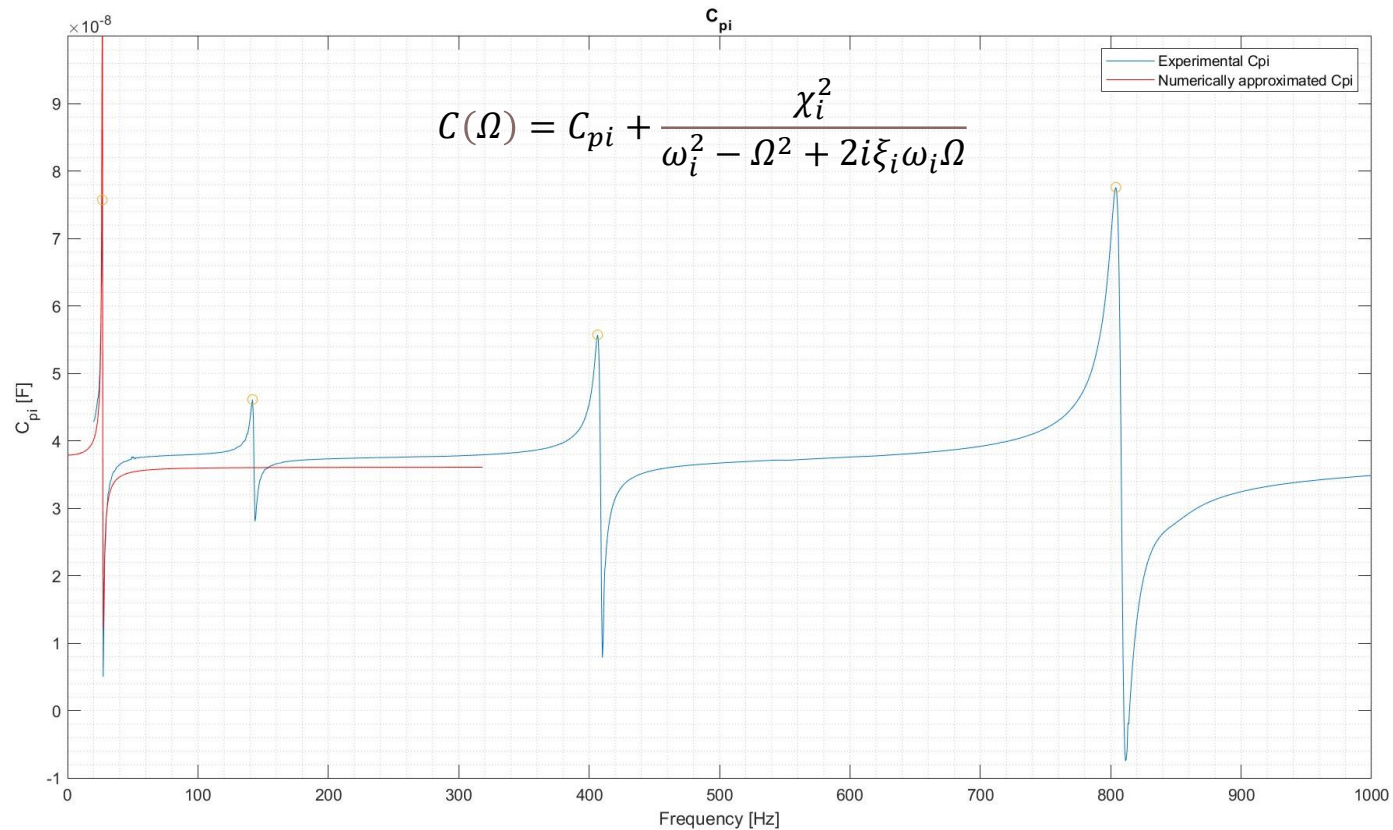
Optimal Values Estimation – C_{pi} & C_0 Estimation



$$C_0 = 4.10 \times 10^{-8} F$$

$$C_{pi} = 3.78 \times 10^{-8} F$$

Optimal Values Estimation – C_{pi} & C_0 Estimation



$$C_0 = 3.79 \times 10^{-8} F$$

$$C_{pi} = 3.61 \times 10^{-8} F$$

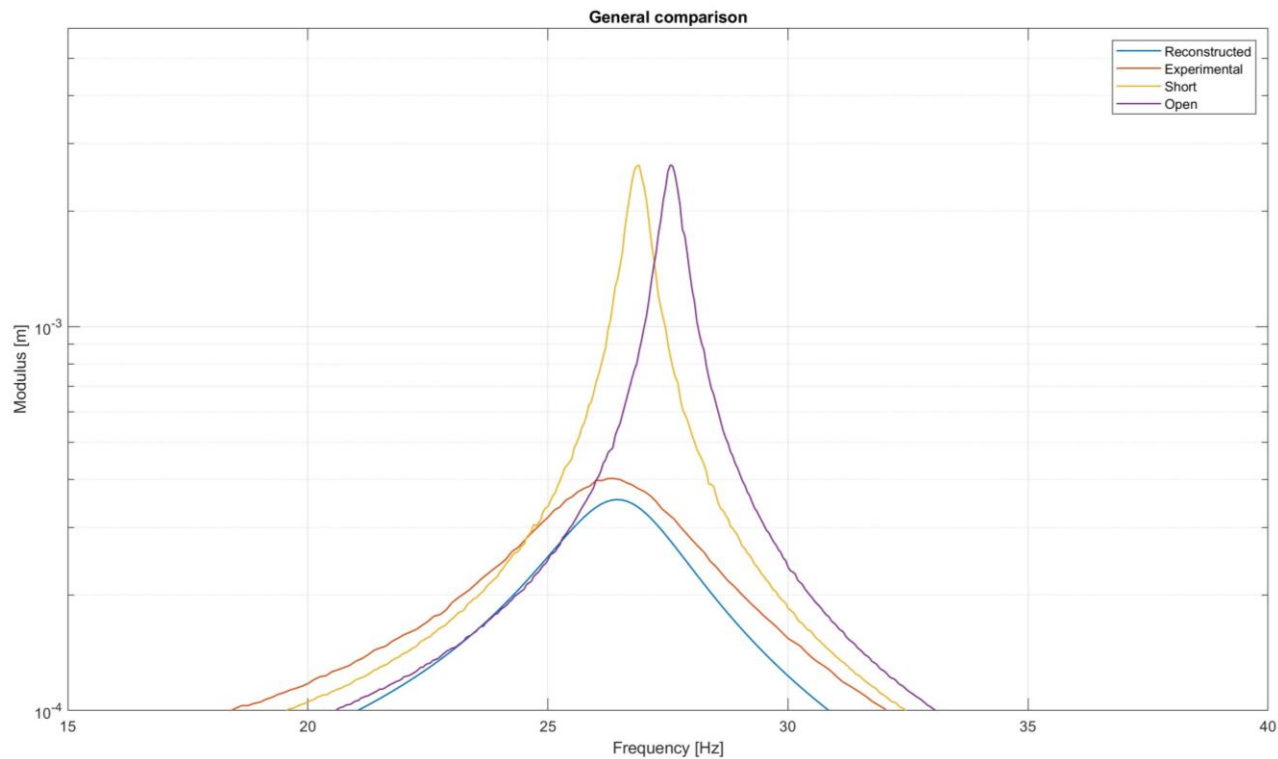
Circuit's components values

	$R_{SH}[k\Omega]$	$R_1[k\Omega]$	$R_s[k\Omega]$	$C_N[nF]$	$C_N^{lim}[nF]$
Taken on field	47.74	7.12	804.79	54.00	39.65
Post processing	49.99	7.46	843.50	51.56	37.85

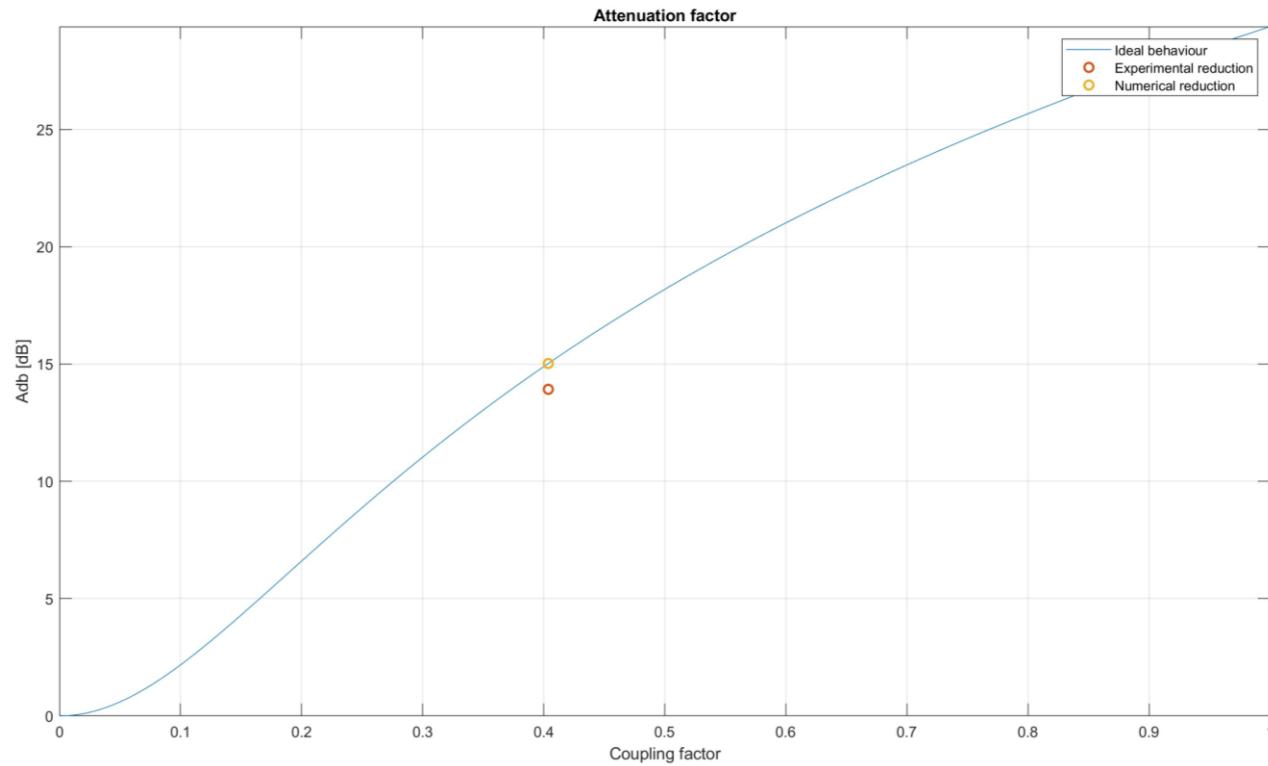
	$R_2[k\Omega]$	$R_{eq}[M\Omega]$	$\hat{R}[k\Omega]$	$\hat{C}[nF]$	β_2
Fixed values	8.1	-50	910	47	0.7

Ideal Response vs Real Response

$$H_{NC}(\Omega) = \frac{1 + i\tau_{opt}\Omega}{(\omega_1^{SC})^2 - (1 + 2\xi_1\omega_1\tau_{opt})\Omega^2 + i\Omega[\tau_{opt}(\omega_1^{OC})^2 + 2\xi_1\omega_1 - \tau_{opt}\Omega^2]}$$



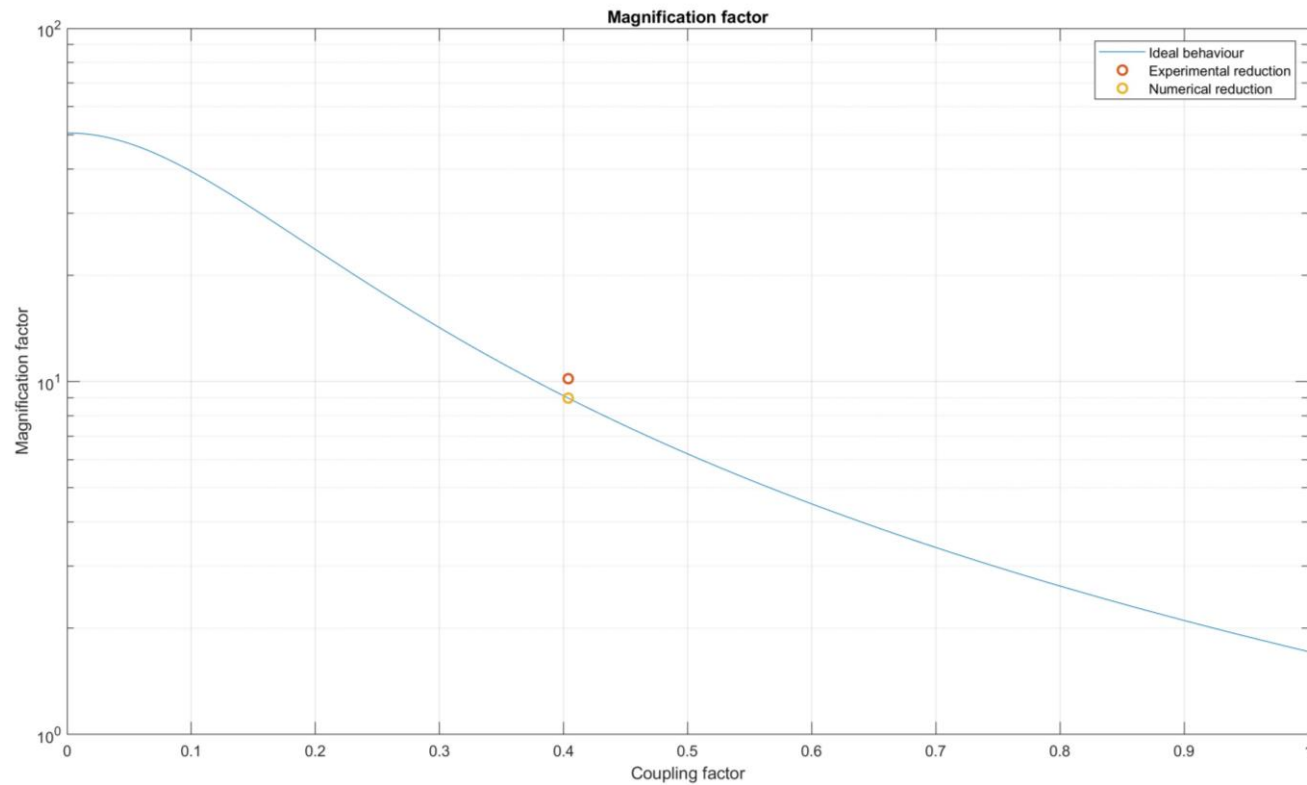
Performance Evaluation



Experimental reduction= 13.92 *dB*

Numerical reduction= 15.02 *dB*

Performance Evaluation – cont'd



Experimental = 10.18

Numerical = 8.97

Thanks for your
attention!